

Steve Crampton

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Work Experience

Software Development Engineer, 2010–present

Amazon.com, Seattle, WA

- Design and implement a new system to generate and manage social-media ads
 - Write and test distributed workflows using SWF (Simple Workflow)
 - Write DAOs for DynamoDB
 - Design and implement systems to integrate between the Facebook Ads API, DynamoDB, Oracle databases, and internal Amazon systems
 - Write unit and integration tests, loggers, and metrics
 - Draft and present design documentations for requirements, systems, and data schemas
- Designed and helped build a new fee-computation system for the Amazon Associates Program, replacing a 10-year old system with a new distributed architecture to provide a more scalable and reliable system
 - Wrote a component to batch up items from queues (internal versions of SQS) and persist them in encrypted form on S3
 - Wrote portions of five other components, including mappers and reducers for Amazon's EMR Hadoop clusters
 - Acted as Scrum Master for team divided between the U.S. and India for the project
- Wrote Hive jobs for, maintained, and optimized an in-house Hadoop cluster to perform view-through attribution for offsite-display ads
- Designed and implemented system expansions of the Associates Program to new marketplaces with different languages and currencies
- Extended and maintained ad-serving and tracking systems for the Amazon Associates Program
- Designed and ran ETL jobs and SQL queries on large databases to obtain relevant, actionable information for decisionmakers
- Use Linux tools to parse logs and gather metrics for operational analysis
- Troubleshoot and resolve operational issues related to a wide array of services supported by the Global Traffic Platform
- Interview job candidates, write job descriptions, and otherwise participate in hiring decisions

Computer Science Instructor, 2010

ITT Technical Institute, Jacksonville, FL

- Taught Introduction to the Linux Operating System and Linux System Administration

Software Developer, 2009

U.S. Geological Survey, Hawaii National Park, HI

- Determined system requirements and drafted proposal
- Designed algorithm to track multiple objects flying in front of a video camera
- Built portable code to run on Windows, MacIntosh, and Linux systems
- Designed a user-friendly interface and documented the project to client specifications

Work Experience (continued)

Technology and Math Teacher, 2008–10

Kirby-Smith Middle School, Jacksonville, FL

- Designed and taught electives and math classes in a dedicated math and science magnet school
- Created a thin-client computer lab using Linux and other free, open-source software
 - Drafted a standards-based curriculum proposal to support the lab
 - Secured funding from school and district leaders
 - Designed and constructed the physical and electrical layout
- Configured and maintained a Moodle course-management website with MySQL and Apache

Computer Science and Technology Teacher, 2004–08

Dwight-Englewood School, Englewood, NJ

- Designed and taught AP Computer Science and other technology courses
- Advised independent student projects on generating functions, chatterbots, computational fluid dynamics, and 3D animation

Computer Science Lecturer, 2000–04

Boston University, Boston, MA

- Taught Introduction to Computers to sections as large as 120 students
- Supervised 2 teaching fellows and 4 graders

Political Science Lecturer, 1998–99

Salem State University, Salem, MA

Professional Actor, Singer, Music Teacher, and Music Director, 1995–99

Various Theaters and Schools in the Boston, MA area

Associate Attorney, 1989–94

Dow, Lohnes and Albertson, Washington, DC

- Delivered arguments, interrogated witnesses, negotiated settlements, and advised clients
- Drew up letters, pleadings, briefs, memoranda, affidavits, and other legal documents
- Supervised other attorneys, legal assistants, and paralegals

Skills and Interests

Computer Projects Portfolio

- <http://www.stevec.org/computer.html>

Computer Skills

- **Languages:** C, C++, Python, Java, LISP, Linux/Mac/Windows Scripting, Objective-C, VB
- **Cloud Services (Amazon Web Services):** Simple Workflow (SWF), Simple Storage Service (S3), DynamoDB, Elastic Compute Cloud (EC2), Elastic Map Reduce (EMR), Virtual Private Cloud (VPC), Simple Queue Service (SQS)
- **Databases:** DynamoDB, MySQL, Oracle, SQL Server
- **Web Server:** JSP, HTML, CSS, JavaScript, PHP, Apache, Tomcat
- **Office Applications:** OpenOffice, Word, Excel, Access, PowerPoint
- **Multimedia:** Audacity, AudioDesk, Garageband, Photoshop, G.I.M.P., Blender, iMovie, ffmpeg, ProTools, Windows Movie Maker

Education

M.A. in Computer Science, 2004

Ph.D. Student, 2000-04

Boston University, Boston, MA

- **Chancellor's Award**, Boston University's Science and Technology Day 2003
- **Coursework:** computer vision, computer graphics, networks, complexity, algorithms, systems, languages, algebra, multivariable calculus, probability, statistics, stochastic processes, logic, and discrete mathematics
- Published in the fields of human-computer interaction and computer vision

J.D., with honors, 1989

George Washington University Law School, Washington, DC

- **First Place**, Van Vleck Moot Court Competition
- **Member**, *The George Washington Law Review*

B.A., cum laude, in Mathematics and Computer Science, 1986

Middlebury College, Middlebury, VT

- **Senior Thesis:** The Data Encryption Standard

Professional Development

Publications

- M. Betke, D. E. Hirsh, N. C. Makris, G. F. McCracken, M. Procopio, N. I. Hristov, S. Tang, A. Bagchi, J. Reichard, J. Horn, S. Crampton, C. J. Cleveland, and T. H. Kunz. "Thermal Imaging Reveals Significantly Smaller Brazilian Free-tailed Bat Colonies than Previously Estimated." *Journal of Mammalogy*, 89(1), February 2008.
- S. Crampton, "Counting Fingers in Real Time Using Computer-Vision Techniques," Masters Thesis, May 2004, Boston University.
- D. E. Hirsh, M. Betke, S. Crampton, J. Horn, and T. H. Kunz, "Censusing Brazilian Free-tailed Bats Using Infrared Thermal Imaging and Computer Vision Methods," 33rd Annual North American Symposium on Bat Research, NASBR, Lincoln, Nebraska, October, 2003.
- S. Crampton and M. Betke, "Counting Fingers in Real Time: A Webcam-Based Human-Computer Interface with Game Applications," *Proceedings of the Conference on Universal Access in Human-Computer Interaction*, affiliated with HCI International, pp. 1357–1361, Crete, Greece, June 2003.
- S. Crampton, M. Betke, T. Kunz, and J. Horn, "A Prototype Computer-Vision System for Tracking Bats," Presented at the Bat Project at NCEAS, Santa Barbara, CA, January 2003.
- S. Crampton and M. Betke, "Finger Counter: A Human-Computer Interface." 7th ERCIM Workshop "User Interfaces for All," UI4ALL 2002, pp. 195–196, Paris, France, October 2002.

Conferences and Workshops

- Presented "Using and Teaching Blender: A 3D Modeling Software" and "Digital Photo Editing with G.I.M.P." at the *International Society for Technology in Education*, Dever, CO, 2010
- Presented "Using and Teaching Blender: A 3D Modeling Software" at the *Florida Educational Technology Conference*, Orlando, FL, 2009
- Presented "3D Modeling and Animation Using Blender" at the *International Society for Technology in Education*, San Antonio, TX, 2008
- JAVA Workshop for AP Computer Science Teachers, *Carnegie Mellon University's School of Computer Science*, Pittsburgh, PA, 2007
- National Educational Computing Conference, *International Society for Technology in Education*, Atlanta, GA, 2007
- Teacher Development Workshop: Computer Gaming, *New Jersey Institute of Technology College of Computing Sciences*, Newark, NJ, 2006
- Fractal Workshop, *Yale University*, New Haven, CT, 2006
- Geographical Information System Workshop, *New Urban Research*, Denver, CO, 2006
- Advanced Adobe Photoshop CS2 Class, 2005, *School of Visual Arts*, New York, NY
- What's New in ArcGIS, *Environmental Systems Research Institute*, New Brunswick, NJ, 2005
- New Jersey Association of Independent Schools Annual Conference, *Newark Academy*, Livingston, NJ, 2004

Organizational Affiliations

- Middle School Officer, 2009–10, *ISTE Special Interest Group for Computing Teachers*
- Communications Officer, 2007–09, *ISTE Special Interest Group for Computing Teachers*
- Member, *International Society for Technology in Education and Computer Science Teachers Association*

Online version can be found at <http://www.stevec.org/resume.html>